

Design and Development of the Skimmer Boat to Overcome Waste in Rivers

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1. Introduction

Rivers are highly diverse and productive ecosystems, contributing to economic growth, food security and human well-being. Globally, an estimated 2 billion people rely directly on rivers for their drinking water and 500 million people (approximately one in 14 people on Earth) live on deltas that are sustained by sediment from rivers, according to the World Wildlife Fund (WWF) [1].

Meanwhile, rivers provide some of the world's most productive fisheries and livelihoods for 60 million people, 55 percent of whom are women. At least 12 million tons of freshwater fishes are caught each year (accounting for some 12 percent of the world's entire fish catch) – this is sufficient to provide protein for at least 160 million people, but very few decision-makers fully appreciate this value of freshwater fish. This is due to a lack of understanding or measurement of the extent to which this supports low-income communities or boosts economies.

Rivers are a very important resource for human life and other living things. The main causes of river pollution are industrial waste, household waste, agricultural waste, and mining activities[1]. Even the Citarum River, the longest river in West Java, was once named the dirtiest river in the world in 2018 [2, 3]. According to Badan Pusat Statistik (BPS), around 46% of rivers in Indonesia are heavily polluted, 32% are in the middle of moderate/heavily polluted, 14% are moderately polluted, and 8% are lightly

polluted [4]. The current situation of rivers is causing ecosystem damage, polluting drinking water sources and threatening biodiversity in Indonesia.

All of these things are nothing else than the results of human activities. For this reason, We are moved to help clean up rivers in Indonesia from waste. From year to year the government has tried various ways to overcome the large amount of waste in the river, real examples such as Interceptor 001, Trash Boom, Arister, Trash Conveyor, and others, these tools aim to clean the river from the trash mixed into it [5, 6]. Although these tools are effective in cleaning dirty rivers in Indonesia, these tools have several serious problems that prevent them from carrying out their goals optimally. One of the most obvious problems is the high production & labor costs, so they cannot be distributed optimally [7, 8].

The objective of our research is to focus on the environmental impact, required materials, and the system that will be utilized in our project.

2. Method and Experimental Details

This research will be conducted from October to November 2024. With details of 10 days to collect data, 5 days to analyze and compile data and 15 days to finalize and check the results of the work. The research will be conducted by collecting data from digital sources by maximizing the time listed above. Researchers will collect data from information sources, namely journals, papers, and theses. Researchers also prepare models or products as a solution to the problems that have been listed in the background, namely the problem of waste in rivers. Researchers plan to make a multifunctional mini skimmer ship as a solution to the problem of waste in rivers.

2.1 Materials Needed

1. Round Can
2. 1m x 5m net
3. 2 dynamos
4. Mini propeller
5. Gear
6. Battery
7. Electrical circuit using ready-to-use modules, namely the Motor Driver Module with Arduino.
8. Buoyant spheres

2.2 Floating formula

This project can float using *Archimedes' Principle*. *Archimedes' Principle* is a law that states that every object that is partially or completely submerged in a liquid, or some liquid, has an upward force on the object, or what is often called buoyancy.

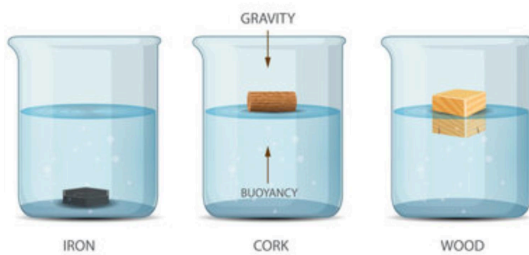
A. Archimedes' Principle Formula

In simple form, the Archimedes law states that the *buoyant force* on an object is equal to the weight of the fluid displaced by the object. Mathematically written as:

$$F_b = \rho \times g \times V_{\text{submerged}}$$

Where F_b is the buoyant force, ρ is the density of the fluid, V is the submerged volume, and g is the acceleration due to gravity.

ARCHIMEDES PRINCIPLE



B. Archimedes' Principle Derivation

We know that the density is defined as

$$\text{Density } (\rho) = \text{Mass } (M) / \text{Volume } (V)$$

Therefore, the mass of the displaced liquid can be written as follows:

$$\text{Mass } (M) = \text{Density } (\rho) \text{Volume } (V)$$

Now, the weight of the displaced liquid can be calculated as follows:

$$\text{Weight} = \text{Mass} \times g = \rho \times V \times g$$

From Archimedes' principle, we know that the apparent loss of weight is equal to the weight of the water displaced therefore the thrust force is given by the following equation:

$$\text{Thrust Force} = \rho \times V \times g$$

Where ρ is the density of the liquid, V is the volume of liquid displaced and g is the acceleration due to gravity.

The thrust force is also called the buoyant force because it is responsible for objects floating. Thus, this equation is also called the law of buoyancy.

C. Drawbacks of Archimedes Principle

1. Ideal Fluid Assumption:

Archimedes' principle is based on the assumption of an ideal fluid with no viscosity or turbulence. In real-world scenarios, fluids often exhibit viscosity and other complexities that can affect the accuracy of calculations.

2. Non-Uniform Fluid Density:

In cases where the density of the fluid varies significantly with depth (stratified fluid), Archimedes' principle might not provide accurate predictions of buoyant forces. This is particularly relevant in oceans and large bodies of water.

3. Compressibility Effects:

Archimedes' principle assumes that the fluid is incompressible. In situations involving gasses or fluids under high pressure, compressibility effects may influence the accuracy of calculations.

4. Irregular Shapes:

The principle assumes that the displaced volume is equal to the volume of the submerged object. However, for irregularly shaped objects or those with complex geometries, accurately determining the volume can be challenging.

5. Surface Tension Effects:

In cases where surface tension is significant, such as very small objects or thin films on a liquid surface, the forces at play can deviate from the ideal conditions assumed by Archimedes' principle.

6. Fluid Acceleration:

The principle does not account for the effects of fluid acceleration or turbulence, which can be relevant in scenarios involving rapidly moving fluids.

7. Non-Buoyant Objects:

Archimedes' principle is primarily concerned with buoyancy and the behavior of objects that are either floating or submerged. It may not be directly applicable to

objects that are entirely outside a fluid or those in contact with a fluid but not submerged.

8. Influence of External Forces:

External forces, such as currents, waves, and winds, can affect the behavior of objects in fluids, and these factors are not explicitly considered by Archimedes' principle alone.

9. Limited to Static Equilibrium:

Archimedes' principle is most applicable when an object is in static equilibrium (not accelerating). It might not fully describe the behavior of objects in dynamic or changing fluid conditions.

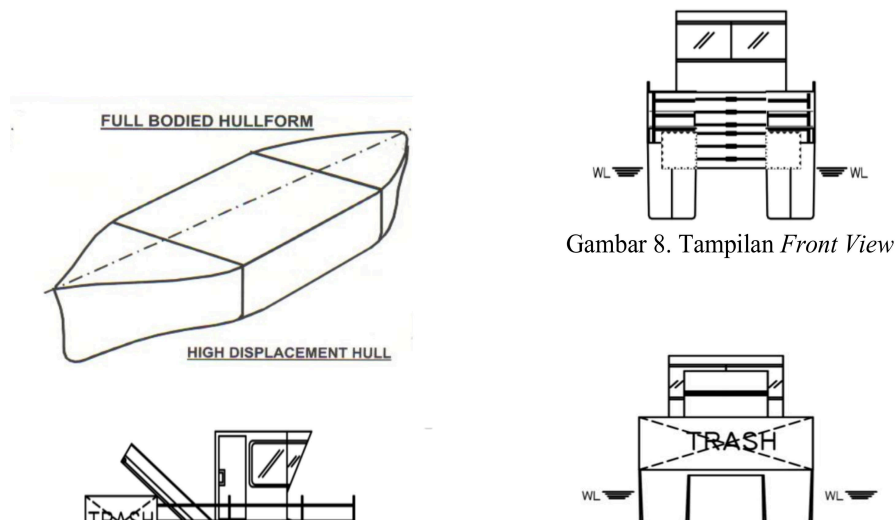
D. Conclusion

Archimedes' principle is a fundamental concept in fluid mechanics with wide-ranging applications. From shipbuilding to submarine technology and even recreational activities such as scuba diving, the principle continues to influence a wide range of engineering and scientific endeavors. By understanding the mechanics of buoyancy, we gain valuable insights into the behavior of objects in fluids, making it an indispensable tool for a variety of practical applications.

2.3 Boat System Details

A suitable ship for floating has a block or concave shape with an air cavity inside. This air cavity makes the ship have a density that is lighter than air, so it can float. We conclude that the body of the skimmer ship will be round shaped.

For any vessel operating in open water, payload must be traded off, to some extent, against sailing performance. The hull can be thought of as three separate sections that converge along the length of the vessel. The fore and aft sections taper towards the bow and stern while the parallel body section amidships is rounded. The proportion of this central region to the total underwater length varies from one design to another.



Gambar 8. Tampilan *Front View*

A. Propulsion System

The propulsion system we use is quite similar to how an RC boat works. Radio-controlled (RC) boats work by using a transmitter to send signals to a receiver on the boat that then relays the signals to the boat's control system.

A transmitter, a handheld device with buttons and a joystick that sends signals to the boat. A receiver mounted on the boat that receives signals from the transmitter. An electric motor that powers the boat. A steering servo mounted on the boat's rudder that receives signals from the receiver and moves the rudder. A battery provides power for the motor and other components.

The RC boat's turbine produces power that drives the gearbox, which then powers the propeller. The turbine may have two sets of rotating blades that transfer power to the output shaft and the outboard drive. The propeller blades create a large surface area for pressure to push on. The front of the blades is pulled by suction, while the back is pushed by higher pressure. This combination of forces produces forward thrust.

B. Floating System

The skimmer vessel will be equipped with buoyant spheres, attached to its underside with waterproof adhesive. These spheres will contribute to the vessel's buoyancy and maneuverability. The choice of spheres was influenced by their lightweight properties, small footprint, and minimal obstruction of the waste capture mechanism.

The buoyant spheres will be attached to the underside of the vessel at intervals of 6 centimeters, forming a ring around the waste collection net. A total of five spheres will be utilized. Due to the vessel's center of gravity being located at its center, the buoyant spheres will be concentrated in this central region.

C. Trash Receiving System

Our vessel's waste collection system will utilize a right-angled trapezoidal conveyor belt. The trapezoidal shape is chosen to optimize the entry of collected waste into the net, ensuring minimal disruption. The conveyor belt will transport floating debris on the water's surface. Furthermore, the conveyor belt option is selected for its minimal environmental impact compared to traditional nets, which can disrupt aquatic ecosystems.

The conveyor belt system will utilize an electric motor and bicycle chain to drive the belt's rotation. The belt itself will be made of PVC.

D. Net

Nets play an important role in transporting waste in mini skimmer boats. The nets catch and hold waste floating on the water surface such as plastic, leaves, among other light materials. Making waste collection easier because the nets are able to trap materials carried by the current.

E. Sound Sensor

We use motion sensors to detect trash storage when it is almost full. makes it easy to know if trash storage is full. When the storage is full there will be an alarm that can be seen from the alarm application used. Motion sensors work by detecting changes in infrared radiation or heat, and movement within their range. Motion sensors can trigger connected devices, such as security cameras or alarms, when they detect movement.

3. Result and Discussion

The expected outcome of this project is to help and motivate the residents around the river by considering some of the project's limitations in terms of relatively small size, limited capacity, and available energy, we conclude that small rivers or tributaries can be an ideal choice for this project. These rivers usually have calm currents, so that the tools that work are not easily carried away by the current and can more effectively trap garbage or settle particles.

Some of the main reasons for choosing a small river are [9]:

1. Calmer currents: Small rivers generally have calmer currents, making it easier for tools to process incoming waste.
2. Less Water Volume: With a smaller water volume, tools can be more
3. efficient and effective in cleaning the river because it is easier to handle the existing waste.
4. Measurable Environmental Impact: In the event of a tool malfunction, the level of impact on the ecosystem of a small river compared to a large river is smaller; this provides a safer space for repairs.
5. Easier Accessibility: Small rivers are usually more accessible; making it easier to monitor and maintain tools regularly.

In general, creeks or small rivers provide a more controlled and safe environment when testing equipment to clean up trash.

4. Conclusion

4.1 the success of previous inventors

The success of using waste cleaning tools in Indonesia is influenced by several factors, such as appropriate technology, community involvement, government support, multi-party cooperation, and periodic evaluation and improvement [10]. For example, Interceptor 001, this tool is capable of cleaning up to 10 tons of plastic waste and has succeeded in reducing 70% of plastic waste in several rivers in Indonesia [11]. Another example is the River Trash Cleaner (RTC). This tool was developed by Indonesian students and is designed to automatically clean waste at river mouths [12].

4.2 suitable water terrain

After Understanding the Limitations of the project such as the relatively small size, limited capacity, and energy available. We can conclude that the ideal small river or tributary terrain for the use of this project is a river with a relatively calm current. We chose a small river/tributary because it has a calmer water flow.

Small rivers generally have slower currents than large rivers, so the tool is not easily carried away by the current and can work more effectively in trapping waste or settling particles. Smaller water volume. With a smaller volume of water, this tool can more easily process incoming waste. This allows the tool to work more efficiently and effectively in cleaning the river. More measurable environmental impact. If there is a malfunction or problem with the tool, the impact on the ecosystem of the small river will be more limited compared to a large river. This provides a safer space to make repairs. Easier accessibility. Small rivers are usually easier to access and reach, making it easier to monitor and maintain the tool regularly. In short, small rivers or streams offer a more controlled and safe environment to test waste cleaning tools, so they can be a good first step in learning and applying science to solve environmental problems.

4.3 Conclusion

Our project is innovative due to its cost-effectiveness and ease of fabrication. The use of affordable materials ensures that this device can be constructed by a wide range of individuals. Additionally, its functionality and simplicity make it an ideal tool for teaching environmental conservation through practical, miniature models.

5. Acknowledgement's a picture

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